

CSC10011 Software Engineering for AI-enabled Systems

Project Assignment 3 (PA3)

Due date: 4/6/2025

Total score: 75 points

This is a project assignment which will be submitted by each team.

You have the following documents to be done and submitted.

a – Revise SAD (15 points)

Your team is asked to revise the Software Architecture Document (SAD) that you submitted in PA2. Your team needs to update Sections 1-4 and fill the content for Sections 5 and 6, based on the feedback for PA2 from the TA's and more information you have about your project now.

For Section 5, draw a deployment diagram(s) using UML and describe briefly each node. Note that the deployment diagram in UML consists of nodes that run your system's components and sub-systems and links among them. For example, if your system runs only on Android devices, without servers, then the diagram has only one node. If your system is a Web-based application, it typically has to run on one or more Web servers, desktop-based Web clients, and mobile Web clients, so the deployment diagram should have three nodes connecting each other via Internet (either HTTP or HTTPS protocols).

For Section 6, provide structures for folders that store your code and supporting files. For example, you have a Web server, provide a folder structure for source code and supporting files on your Web server. Let's just focus on the main folders only. The folder structure can look like a folder structure on Windows.

b – ML model training, deployment and evaluation (20 points)

In the SAD document, provide details on how the ML models chosen are deployed and operated when the software is deployed. Are ML models are deployed dynamically or statically? How ML models are retrained, fine-tuned or revised, evaluated and redeployed back to serve users.

In this PA, your team is also asked to do the following and include the results in the SAD document:

- Find/choose/collect datasets needed to evaluate, and describe the datasets.
- Define metrics for evaluating your models, acceptance threshold for each metric (for example acceptance accuracy of 80% is expected).

(Details on model evaluation will be provided in PA4)

c – UI prototype (15 points)

Your team needs to sketch the user interface for the system your team is designing. You can use tools such as Figma, Claude Design, Google Design, Paper Design, and Pencil Dev to design the UI. You need to sketch the UI for the main screen and at least 5 screens for the key scenarios. The key scenarios must follow what you describe in the use-case specifications.

For each screen, you need to describe its purpose, what it shows and how users can use it. To do so, you can copy all screens to MS Word and provide the description below each screen.

d – Working software (20 points)

Implement at key features of the application, focusing on AI-enabled features. At least two main flows of two main use-cases are working. To report the working software, (1) include a readme file describing what features have been implemented, limitations and known defects, (2) prepare a video capturing your demo. The video lasts between 2 to 3 minutes.

c – Weekly report (5 points)

Prepare weekly reports for the weeks starting from the last reported week until the time of submission. Fill out the template with all information required.

Submission Guidelines

You must copy all documents to the directory named **PA3-Group[GroupId]** and compress the whole directory to the zip/rar file named PA3-Group[GroupId].zip/.rar. For example: **PA3-Group01.zip**

Grading Criteria

- Writing: easy to understand, few errors.
- Well formatted texts, diagrams, charts.
- Level of completeness of the documents, based on the information available at this time.
- The software architecture must capture key components necessary for the system.
- The UI design must match the use-cases modeled in the previous assignments.
- The UI has sufficient elements (labels, controls, etc.) for users to complete their tasks.
- Details on how ML models are deployed and operated are clearly given.
- Details on how the datasets are collected/chosen, their key statistics (size, how many data points, etc.)
- Provide initial results obtained from the model evaluation.
- Working software with at least two use cases working, demonstrated in the video.